WEEK 1 • PROJECT 1 • COMPULSORY

SplitSmart

Dart Expense Splitter | Beginner to intermediate

Build a Dart program that tracks expenses shared by a group and calculates how much each member owes or should receive. No Flutter UI is required.

Skills: functions, collections, classes, enums, validation, exception handling, and problem solving.

Required Features

- Add group members with unique names or IDs.
- Record expenses with a title, positive amount, payer, category, and selected participants; split the amount equally among participants.
- List expenses and filter them by member or category.
- Calculate each member's total paid, share owed, and net balance across multiple expenses.
- Handle duplicate members, unknown members, invalid amounts, and empty participant lists with clear errors.

Two Accepted Ways to Run the Project

Local Dart: build a terminal menu with dart:io input and run your file using `dart run main.dart`.
DartPad fallback: supply data through method calls in main() and print the results. Both routes meet the core requirement; interactive input is not required for the DartPad route.

Example usage to implement (ExpenseGroup and its methods must be written by you):

```
void main() {
  final group = ExpenseGroup();
  group.addMember('Asha');
  group.addMember('Ravi');
  group.addMember('Neha');
  group.addExpense(
    title: 'Dinner', amount: 900, paidBy: 'Asha',
    category: 'Food',
    splitBetween: ['Asha', 'Ravi', 'Neha'],
  );
  group.printBalances();
}
```

Expected: Asha should receive INR 600; Ravi and Neha each owe INR 300. Demonstrate additional scenarios with multiple payers. For the basic version, use whole-rupee expenses that divide evenly.

Optional Extensions

Suggest settlement transfers; support unequal splits; edit/delete expenses; handle rounding consistently; save/load JSON files locally; return summaries as records.

Resources

Dart language guide

Running local Dart files

DartPad and supported libraries